Build & Evaluate a Linear Regression Model

House Price Predictor - California Housing Dataset

Project Report

This report summarizes a small machine learning workflow implemented in Jupyter Notebook. The project uses the California Housing dataset, trains a Linear Regression model, and evaluates the predictions using common regression metrics.

Objective

The objective of this task is to understand the complete ML workflow: loading data, exploring the dataset, splitting into training and test sets, training a regression model, evaluating performance, and presenting the results.

Dataset

The California Housing dataset contains housing-related features such as median income, house age, average rooms, average bedrooms, population, and geographical location. The target variable is Median House Value.

Model Evaluation

Metric	Value
Mean Absolute Error (MAE)	0.533
Root Mean Squared Error (RMSE)	0.746
R-squared (R2)	0.576

Workflow Summary

- Imported pandas, NumPy, Matplotlib, and scikit-learn.
- Loaded the California Housing dataset and combined features with the target column.
- Split the data into training and testing sets using an 80/20 split.
- Trained a LinearRegression model.
- Predicted on the test set and calculated MAE, RMSE, and R2.
- Plotted Actual vs Predicted values to visualize performance.

Conclusion

The model captures the general trend in house prices, but the scatter plot shows noticeable spread around the ideal diagonal line. This suggests there is room for improvement through feature engineering, stronger regression models, and preprocessing steps.

Deliverable Note

This PDF is prepared as a clean submission-ready summary for the task deliverable.